MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 12AWP12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Magnetic deflection, electrostatic focus, aluminized screen

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb2	Ehk	Rg1	Eb1
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	Vdc
Maximum:	6.9	0	770	13,000	+180	Note 1	1,100
Minimum:	5.7	-180	---	7,000	-180	---	-550
TEST CONDITIONS: (See note 2)	6.3	Adj	300	10,000	---	---	Focus

GENERAL:

Qualification - Required

Ⓑ denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:33Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS	
					MINIMUM	MAXIMUM
	Qualification inspection					
5201	Electrode current (grid No. 2)	-	Ec1 = 0	Ic2	-15	+15
5216	Cathode illumination	-		---	---	---
1331	Direct-interelectrode capacitance	-	Cathode to all Control grid to all	Ck Cg1	--- ---	8.0 10.0
1141	Pressure (implosion)	-		---	---	---
5111	Vibration	-		Width	---	2.0
	Quality conformance inspection, part 1					
5201	Voltage breakdown	-		---	---	---
5201	Voltage breakdown (magnetic types)	-		---	---	---
5206	Gas ratio	6		GR	---	0.25
5106	Bulb, screen, and faceplate quality	-		---	---	---
5223	Modulation	4,5	Ib2 = 25 µAdc; overscanned de-focused pattern	ΔEc1	---	20
5231	Spot position	-		---	---	0.625
---	Cathode emission	-		---	100	---
5241	Grid cutoff voltage	-		Ec1	-15	-45
(B) 5101	Neck straightness	3		---	---	---
5251	Grid No. 1 leakage current	-		---	---	---
	Quality conformance inspection, part 2					
1301	Heater current	-		If	270	330
5101	Side terminal and base alinement	-	Halfway between pins 1 and 9	---	---	---
5101	Neck and bulb alinement	-		---	---	---
5101	Face tilt	-		---	---	---
(B) 5240	Focusing voltage	-	Ib = 25 µAdc	Eb1	0	450
5216	Stray light emission (conventional types)	-	Eb2 = 13,000 Vdc; Ec2 = 770 Vdc; Eb1 = 0	---	---	---
5221	Screen	4,5		---	---	---
---	Fluorescent brightness	5		cBf	625	---
5226	Line width A	-	Ib2 = 25 µAdc	Width	---	0.45
5251	Heater-cathode leakage current	-		---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:33Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS	
					MINIMUM	MAXIMUM
	Quality conformance inspection, part 2 -Continued					
5251	Grid No. 2 leakage current	-		---	---	---
Ⓑ 1105	Permanence of marking	-		---	---	---
	Quality conformance inspection, part 3					
---	Life test	-	Group C; Eb2 = 13,000 Vdc; Ec2 = 770 Vdc; Eb1 at focus; Ib2 = 5 µAdc; t = 500 hours (min)	---	---	---
---	Life-test end points:					
5226	Line width A	-	Ib2 = 25 µAdc	Width	---	0.45
5223	Modulation	-	Overscanned de-focused pattern	Ec1	---	20

NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed 0.5 Meg. When Ec2 is less than 330 Vdc, Rg1 shall exceed 1.5 Meg.
2. All tests except screen persistence and fluorescent brightness can be made at 50 percent duty cycle pulsi
3. Cylinder shall be 4 inches (101.60 mm) long and 0.901 inch (22.89 mm) maximum inside diameter.
4. After continuous excitation and when equilibrium has been reached, the time constant shall be plotted. B between 0.1 and 0.3 second, the time constant shall be between 75 and 105 milliseconds. The cB light output valu at 0.1 second shall be a minimum of 200 cB greater than the cB light output value recorded at 1.0 second.
5. These recordings shall be made with an Eb2 = 10,000 Vdc and an Ib2 = 5 µAdc, and a 7.1 x 7.1 cm pattern d
6. This test to be performed at the conclusion of the holding period.

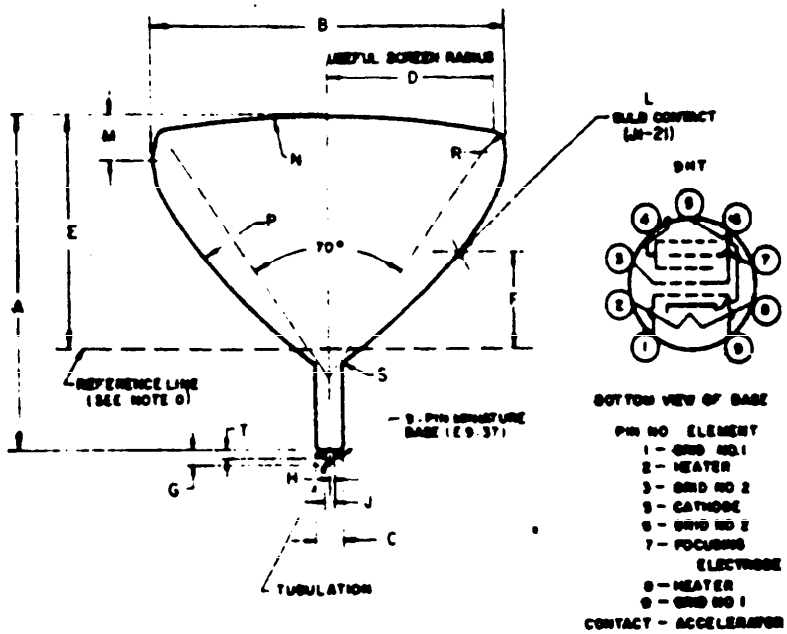
Review activities:
Navy - AS, OS, MC, CG, SH

Preparing a
Navy - EC

Agent:
DSA - ES

(Project 59

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:33Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
L	Bulb contact: J1-21	
K	Base: E9-37	
Quality conformance inspection, part 2		
A	12.063 (306.40)	12.438 (315.93)
B	12.312 (312.72)	12.562 (319.07)
C	.810 (20.57)	.900 (22.86)
D	5.500 (139.70)	---
E	8.375 (212.73)	8.625 (219.08)
F	2.812 (71.42)	3.188 (80.98)
G	---	.438 (11.13)
H	---	.125 (3.18)
J	---	.250 (6.35)
Reference dimensions (see note b)		
M	1.359 (34.52)	
N	40 (1016.00) R	
P	20 (508.00) R	
R	.438 (11.13) R	
S	.125 (3.18) R	
T	.250 (6.35)	

NOTES:

a. Reference line is determined by the point where leading edge of 1.640 inches (41.66 mm) reference line gage will stop (JETEC No. 128).

b. These dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 12AWP12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:33Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:33Z
Check the source to verify that this is the current version before use.